

10.20 The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security j is related to the risk premium on the market portfolio. That is

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

where r_j and r_f are the returns to security j and the risk-free rate, respectively, r_m is the return on the market portfolio, and β_j is the j th security's "beta" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

- Use the observations on Microsoft in the data file *capm5* to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?
- It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \dots, 180$. Does this variable potentially satisfy the conditions IV1–IV3? Create *RANK* and obtain the first-stage regression results. Is the coefficient of *RANK* very significant? What is the R^2 of the first-stage regression? Can *RANK* be regarded as a strong IV?
- Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at the 1% level of significance. Can we conclude that the market return is exogenous?
- Use *RANK* as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?
- Create a new variable $POS = 1$ if the market return $(r_m - r_f)$ is positive, and zero otherwise. Obtain the first-stage regression results using both *RANK* and *POS* as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the R^2 of the first-stage regression?
- Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e). Can we conclude that the market return is exogenous at the 1% level of significance?
- Obtain the IV/2SLS estimates of the CAPM model using *RANK* and *POS* as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?
- Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

(a)

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003250	0.006036	0.538	0.591
x	1.201840	0.122152	9.839	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$$r_{\text{msft}} - r_f = 0.003250 + 1.201840(r_m - r_f)$$

由於微軟的估計係數 $\text{beta_hat} = 1.20$ ，明顯大於 1。這意味著當市場上漲 1% 時，微軟預計會上漲 1.2%；當市場下跌 1% 時，微軟預計會下跌 1.2%。因此，我們將微軟歸類為 Risky，因為它的波動性大於整體市場組合。

(b)

Call:

```
lm(formula = x ~ RANK, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.110497	-0.006308	0.001497	0.009433	0.029513

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7.903e-02	2.195e-03	-36.0	<2e-16 ***
RANK	9.067e-04	2.104e-05	43.1	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared: 0.9126, Adjusted R-squared: 0.9121
F-statistic: 1858 on 1 and 178 DF, p-value: < 2.2e-16

IV1 (相關性)：RANK 的 t 值高達 43.1， P 值趨近於 0，說明 RANK 與內生變數 x 強烈相關。滿足。

IV2 (外生性)：由於 RANK 是根據 x 本身排序產生的，它理論上與模型誤差項不相關。通常在計量實務中，這種排序工具變數被視為外生。滿足。

IV3：RANK 除了透過影響 x 來影響 y 之外，沒有其他管道影響 y 。滿足。

RANK 的係數顯著性： $\Pr(>|t|) < 2e-16$ ，係數在 1% 顯著水準下顯著

且 RANK 可視為 strong IV，因相關性極高： R^2 高達 0.91，代表它與內生變數有極強的關聯；統計檢定量極大： F 統計量 (1858) 遠遠超過了傳統門檻值 10；顯著性： t 值 (43.1) 顯示其統計顯著性非常穩健。因此，在後續的 2SLS 估計中，使用 RANK 作為工具變數是理想的，它能提供一致 (consistent) 且穩定的估計結果。

(c)

v_hat 的係數估計值為 -0.874599，其對應的 p 值為 0.0428，雖然在 5% 顯著水準下顯著，但並未達到題目要求檢定的 1% 顯著水準 ($p < 0.01$)，因此在 1% 的嚴格標準下，市場報酬在統計上仍可被視為外生 (exogenous)。

```
>
> # (c) 殘差法進行內生性檢定 (Hausman Test 手動法)
> v_hat <- residuals(first_stage_b)
> augmented_reg <- lm(y ~ x + v_hat, data = df)
> summary(augmented_reg)

Call:
lm(formula = y ~ x + v_hat, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27140 -0.04213 -0.00911  0.03423  0.34887

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.005984   0.504   0.6146
x            1.278318   0.126749  10.085 <2e-16 ***
v_hat       -0.874599   0.428626  -2.040   0.0428 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672,    Adjusted R-squared:  0.36
F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

(d)

測量誤差產生的偏誤：在計量中，如果自變數（市場報酬）存在測量誤差，OLS 估計量會產生「趨向於零的偏誤」，也就是 OLS 會低估真正的 Beta 值。

IV 的修正作用：工具變數法（IV/2SLS）可以用來修正這種測量誤差產生的偏誤。因為我們觀察到 IV 的估計值（1.278）高於 OLS 的估計值（1.202），這正符合「IV 修正了 OLS 低估現象」的理論預期。

```
> # (d) 2SLS 估計 (使用 RANK)
> iv_model_d <- ivreg(y ~ x | RANK, data = df)
> summary(iv_model_d)
```

Call:
ivreg(formula = y ~ x | RANK, data = df)

Residuals:

Min	1Q	Median	3Q	Max
-0.271625	-0.049675	-0.009693	0.037683	0.355579

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003018	0.006044	0.499	0.618
x	1.278318	0.128011	9.986	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-squared: 0.3508, Adjusted R-squared: 0.3472
Wald test: 99.72 on 1 and 178 DF, p-value: < 2.2e-16

(e)

```
>
> # (e) 增加 POS 工具變數
> df <- df %>% mutate(POS = ifelse(x > 0, 1, 0))
> first_stage_e <- lm(x ~ RANK + POS, data = df)
> summary(first_stage_e)
```

Call:
lm(formula = x ~ RANK + POS, data = df)

Residuals:

Min	1Q	Median	3Q	Max
-0.109182	-0.006732	0.002858	0.008936	0.026652

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.0804216	0.0022622	-35.55	<2e-16 ***
RANK	0.0009819	0.0000400	24.55	<2e-16 ***
POS	-0.0092762	0.0042156	-2.20	0.0291 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared: 0.9149, Adjusted R-squared: 0.9139
F-statistic: 951.3 on 2 and 177 DF, p-value: < 2.2e-16

在同時使用 RANK 與 POS 作為工具變數的第一階段回歸中，F 統計量高達 951.3，遠遠超過傳統判定強工具變數的門檻值 10，且兩者的聯合顯著性極高（p-value < 2.2e-16），因此可以判斷我們擁有足夠強的工具變數；此外，模型的回歸係數決定係數 (R^2) 為 0.9149，顯示這組工具變數能有效解釋市場風險溢酬變數絕大部分的變異。

(f)

Wu-Hausman 檢定結果，其統計量為 4.862，對應的 p-value 為 0.0287。由於此 p-value 大於題目要求的 1% 顯著水準（ $0.0287 > 0.01$ ），因此在 1% 的水準下我們無法拒絕虛無假設，結論仍為市場報酬是外生 (exogenous) 的。

(g)

使用 RANK 與 POS 作為工具變數後，IV 估計出的市場 Beta 值為 1.283118。與 (a) 小題的 OLS 估計值（約 1.2018）相比，IV 估計值較高，這符合測量誤差會導致 OLS 產生趨向於零之偏誤（Attenuation Bias）的理論預期，說明 IV 修正了可能存在的低估問題，結果符合預期。

(h)

Sargan 檢定的 P 值為 0.4549。這代表在 5% 顯著水準下，我們無法拒絕工具變數有效的假設。也就是說，RANK 和 POS 這兩個工具變數選得很好，是有效且具備外生性的。

```
> iv_res <- residuals(iv_model_g)
> sargan_reg <- lm(iv_res ~ RANK + POS, data = df)
> sargan_stat <- nrow(df) * summary(sargan_reg)$r.squared
> p_val_sargan <- 1 - pchisq(sargan_stat, df = 1) # 2個IV - 1個內生變數 = df 1
>
> cat("Sargan 檢定統計量:", sargan_stat, "\nP值:", p_val_sargan)
Sargan 檢定統計量: 0.5584634
P值: 0.45488
> View(capm5)
> |
```